

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP)
Equal Cost MultiPath (ECMP) load balancing, fast convergence, widely used. Uses IP Port 89 on Layer 3. Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR)
Dual Election: Highest interface priority → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same roles, second best candidate
ROUTER OPERATION
– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
– Intra-area change: SPF recalculation
– Inter-area change: No SPF needed — ABR handles updates
– Build the routing table from SPF results
AREAS
– AS split into areas (sub-domains), each with a 32-bit Area ID (e.g., 0.0.0.0 = Area 0)

Backbone Area (Area 0): Core of the OSPF domain (must exist). Must connect to all other areas (directly or via virtual links). Must be contiguous (no disjointed segments). Should not contain end-user networks

Non-Backbone Areas: Connect end users and local resources. All inter-area traffic must transit the backbone

ROUTER TYPES
Area Border Router (ABR): Connects two or more OSPF areas; must have interface in backbone. 1 OSPF DB per Area
Internal Router: All interfaces belong to the same area (non-backbone)
Backbone Router: At least one interface in Area 0. Includes ABRs and routers internal to the backbone
AS Boundary Router (ASBR): Connects to external AS/Network (e.g., BGP). Advertises external routes into OSPF. Can be in backbone or non-backbone area

DESIGN RULES
Rule 1: Backbone (Area 0) must be contiguous — no partitions allowed
Rule 2: Every non-backbone area must connect to Area 0

LSA TYPE OVERVIEW
Type 1 – Router-LSA: Sent by all routers, lists directly connected links (so all outgoing interfaces with state and cost) (intra-area only)
Type 2 – Network-LSA: Sent by DR, lists all routers and DR in broadcast/multi-access network (intra-area only)
Type 3 – Summary-LSA: Sent by ABRs, advertises networks from other areas, flooded in all the areas that are not "totally stubby" (inter-area)
Type 4 – ASBR-Summary: Sent by ABRs, advertises path to ASBRs (inter-area)
Type 5 – AS-External-LSA: Advertises external routes (e.g., BGP); flooded to all non-stub areas – only normal areas
Type 7 – NSSA External: Like Type 5 but used inside NSSA; converted to Type 5 by ABR

AREA TYPES (ALLOWED LSA TYPES)	
Standard (1,2,3,4,5): Normal	Not-So-Stubby Area – NSSA (1,2,3,7):
Stub Area (1,2,3):	(1,2,3,7):
– Blocks external LSAs (Type 5)	– Like stub area, but allows one ASBR inside
– ABR injects a default route (0.0.0.0)	– External routes use LSA Type 7 (converted to Type 5 by ABR)
– Supports LSA Types 1–3	
Totally Stubby Area (1,2):	Totally NSSA (1,2,7):
– Blocks external (Type 5) and summary (Type 3/4)	– Like NSSA but blocks Type 3 so gets default route from ABR
– Only allows default route from ABR	

PACKET TYPES
Type 1 – Hello: Used to discover, maintain, and verify neighbors; forms adjacencies. Also for election of DR,BDR in broadcast networks. Contains network mask(of sending routers' interface), Hello interval(p2p,broadcast: default=10s), Options, Priority(for election), Router dead interval(default=40s), DR/BDR/ID, Neighbors
Type 2 – Database Description (DBD/DD): Exchange summaries of LSAs during adjacency formation (headers only). Contains Interface Max. MTU, Options, I/M/MS bits (Initial, More, Master-slave bit), DD Sequence num, LSA Header
Type 3 – Link State Request (LSR): Sent when a router needs specific LSAs listed in the DBD. Contains Link State Type (router/network), Link State ID, Advertising Router (sender address)
Type 4 – Link State Update (LSU): Used to flood new or updated LSAs. Contains Number of LSAs, full LSAs information
Type 5 – Link State Acknowledgment (LSAck): Confirms receipt of LSAs to ensure reliable flooding. For this you send LSAck or implicitly by sending LSU with same info back. Many acks may be grouped together to a single LSAck.

SUB-PROTOCOLS
Hello Protocol:
– Used for neighbor discovery and parameter negotiation.
– Maintains logical adjacencies on P2P, P2MP, and virtual links.
– Elects DR/BDR on broadcast and NBMA networks.
– Continuously sends hello packets to maintain bidirectional connectivity; failure to receive = neighbor down (in agreed router dead interval at initialization)
Database Sync Protocol:
– Syncs LSDB using Database Description (DBD) packets with only LSA headers.
– Uses I-bit (initial), M-bit (more), and MS-bit (master/slave).
– ExStart: Bi-direction; highest Router-ID = master. Determine initial seq nr
– Exchange: Exchange of DBD packets (LSA headers).
– Loading: Missing LSAs are requested.
– Full: Databases fully synchronized.

OSPF ROUTING AND ECMP
– Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)
– OSPF prefers more specific match (CIDR) and if then still multiple: intra-area > inter-area > external
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing

OSPF ROUTE SELECTION
Intra-Area (O): Source and dest in same area; routes from Type 1 and 2 LSAs
Inter-Area (O IA): Source and dest in different areas within same AS; via Type 3 LSAs through backbone

External (E1/E2): Dest outside AS; info injected by ASBR via redistribution
– E1: Total = external + internal OSPF cost
– E2: Only external cost (default)
Preference order: More specific route > Intra-area > Inter-area > E1 > E2
Cost calculation: Cost = Reference Bandwidth/default 100 Mbps/Interface Bandwidth
IS-IS (IGP)
CLNS – CONNECTIONLESS NETWORK SERVICES
CLNS: ISO Layer 3 datagram service; supports CLNP ES-IS, IS-IS.
CLNP: Connectionless Network Protocol, similar to IP; used in ISO stack (EtherType 0xFFE).
IS-IS: Link-state routing protocol (Layer 3); forms adjacencies with ES-IS; designed for CLNP but extended (Integrated IS-IS) to support IP.
Integrated IS-IS: Allows IP routing with IS-IS; used widely by service providers even without CLNP.

NET ADDRESSING (TYPE OF NSAP ADDRESS)
NSAP: Network Service Access Point
NET = Network Entity Title: Unique router identifier in IS-IS
Format: 49.AAAA.BBBB.BBBB.BBBB.00
49.AAAA...: Area ID (variable length)
BBBB.BBBB.BBBB: System ID (usually 6 bytes = unique router ID)
00: N-selector (NSEL) (always 00 for routers)
System ID: Must be unique per router (often based on lo-IP/MAC)
Example: 49.0001.1921.6800.1024.00 based on IP 192.168.1.24
Area ID: Used for routing hierarchy (like OSPF areas)
IS-IS PACKET TYPES (ALL IN L1 OR L2)
IIH (Hello): Builds and maintains adjacencies; includes system ID, holding time, prio
– Built from 3 functions: discover, build, maintain
– Interval: 10s (default); DIS sends every 3.3s on LANs
– Multipier: Missed Hello limit → Holdtime + interval × Multiplier(default=3)
Multicast:
– 01-80-c2-00-00-14 (All L1ISs)
– 01-80-c2-00-00-15 (All L2ISs)
LSP (Link State PDU): Contains topology info including prefixes with costs; flooded throughout the area → similar to OSPF LSA Type 1
CSNP (Complete SNPP): Sent by DIS; lists all known LSPPs (used for database sync)

PSNP (Partial SNPP): Used to request missing LSPs or acknowledge received LSPs
IS-IS Packet Structure: Common header + TLVs
ROUTER TYPES
L1 Router: Only within one area; no inter-area routing
L2 Router: Backbone router; routes between areas
L1/L2 Router: Acts as both; separates databases, redistributes between levels
BROADCAST/MULTI-ACCESS LINKS: DIS – DESIGNATED IS
– Required on broadcast links (no DIS on p2p)
Sends periodic CSNPs to ensure DB sync, creates pseudonode LSP
No backup DIS in IS-IS

DIS ELECTION
– 1) Highest interface priority (0–127) Cisco default = 64
– 2) Highest SNPA (MAC-Address)
Preamption: Enabled – higher prio router automatically takes ver the DIS Role
POINT-TO-POINT LINKS (NO DIS)
CSNP: Sent once at adjacency startup
LSP: Advertises topology changes (link-state info)
PSNP: Acknowledges received LSPs or requests missing ones
PATH SELECTION
Path Selection Order (in IS-IS) – Lower Metric better:
– L1 intra-area routes
– L2 intra-area routes
– Leaked L2 → L1 (internal metric)
– L1 external (external metric)
– L2 external (external metric)
– Leaked L2 → L1 (external metric)
LEVEL 1 ROUTING
– Intra-area routing only (like OSPF intra-area)
– L1 routers use the closest L1/L2 router for inter-area traffic
L1/L2 routers:
– Do not advertise L2 routes into the L1 area (unless route leaking is active)
– Set Attached bit to signal L2 connectivity to backbone
– L1 routers install a default route to nearest L1/L2
– L1 area like OSPF Totally Stubby Area
– Distribution Bit: Set to 'up' (1) on L2 → L1 leaks; blocks re-advertisement L1 → L2.
– Route-Leaking injects a more specific route into L1 to improve routing

LEVEL 2 ROUTING		
– Routing between areas (inter-area)		
– L1/L2 routers inject L1 routes into L2 topology		
– L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP		
IS-IS VS OSPF		
Feature	IS-IS	OSPF
Layer	L2 (CLNS)	L3 (IP, proto 89)
Encapsulation	No IP, uses TLVs	IP packets
Hello Type	IIH	Hello packet
Area Model	L1/L2	Backbone + Areas
Metric	Cost (default 10)	Cost (bandwidth)
Router ID	System ID (6B)	32-bit Router ID
Adj. Types	L1, L2, L1/L2	DR/BDR, P2P
LSDB	Per level (L1/L2)	Per area
Scaling	Large-scale ISP core	Enterprise/campus
Routing Info	TLVs (flexible)	Fixed LSA types

AUTONOMOUS SYSTEM NUMBERS (ASN)
– Unique ID for each AS; required for Internet routing with BGP

– Private ranges:
– 64512–65535 (legacy 16-bit)
– 4200000000–4294967294 (32-bit)
BGP PEERING / NEIGHBORS
– Two routers with a BGP TCP session (port 179) are called peers or neighbors
– Each BGP router is a BGP speaker
– BGP exchanges routing info between ASes (loop-free, policy-based)
– Supports CIDR, route aggregation; decisions based on policies/rules
PATH ATTRIBUTES
Used for route control and policy enforcement
Well-known mandatory: Always present (e.g., AS-Path, Origin, Next Hop)
Well-known discretionary: Optional but recognized by all (e.g., Local Pref, Atomic Aggregate)
Optional transitive: Passed between ASes (e.g., Community, Aggregator)
Optional non-transitive: Not passed across ASes (e.g., MED, Weight, Originator ID, Cluster ID/List)
NLRI: Routing table info: prefix, prefix length, and associated path attributes
LOOP PREVENTION
BGP uses AS-Path (list of ASNs) to detect loops. If a router sees its own ASN in a received route, it discards it.
AS-Override: Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection
BGP MESSAGES
OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params
– Hold Time: Heartbeat in seconds (default 180s, Cisco), reset by KEEPALIVE/UPDATE; 0 = session down
– BGP Identifier: 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
KEEPALIVE: Sent every 1/3 of Hold Time (default 180s); ensures neighbor liveness (BGP doesn't rely on TCP keepalive)
UPDATE: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
NOTIFICATION: Sent on session error (e.g. hold timer expired); terminates session immediately
BGP NETWORK STATEMENTS
– Purpose: Advertise specific prefixes to BGP peers (does not activate interfaces)
– Prefix must exist exactly in the RIB (from static, connected, or learned route)
– Attributes (e.g., origin, next-hop, MED) depend on how the route exists in RIB
– BGP advertises only the best path for a prefix to peers, even if multiple exist
BEST PATH CALCULATION
– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in RIB; recalculated on:
– Next-hop reachability change
– Interface failure to eBGP peer
– Redistribution change
– Withdrawn path received
– Influence:
– Outbound BGP policy → inbound traffic behavior
– Inbound BGP policy → outbound traffic behavior
BGP BEST PATH SELECTION (IN ORDER):
1) Prefer highest Weight (Cisco-specific, local to router)
2) Prefer highest Local Preference (global within AS)
3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0)
4) Prefer shorter AS path (only length is compared)
5) Prefer lowest origin type: IGP < EGP < Incomplete (I on origin)
6) Prefer lowest MED (Multi-Ext Discriminator) (also called metric)
7) Prefer external (eBGP) over internal (iBGP)
8) For iBGP: prefer path with lowest IGP metric to next-hop
9) For eBGP: Prefer oldest (more stable) path
10) Prefer lowest BGP router ID
11) Prefer path from lowest neighbor IP address
ROUTE FILTERING
– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: prefix-list (IP), filter-list (AS-path), route-map (flexible match/set)
BGP COMMUNITIES
– 32-bit optional, transitive tag (e.g. ASN-value, 65000:100)
– Used to mark routes for policy control across ASes
– Can be added, modified, or removed at each hop
iBGP SCALABILITY
– iBGP does not re-advertise routes between iBGP peers → full mesh required (cause no loop prevention)
– Session count = $\frac{n(n-1)}{2}$ (e.g., 5 routers = 10 sessions, 10 = 45)
ROUTE REFLECTORS (RR)
– Solves iBGP full-mesh scaling by allowing selective route reflection
– Clients only peer with RR; unaware they're clients
– RR Rules:
– From non-client → advertise to clients only
– From client → advertise to all (clients & non-clients)
– From eBGP peer → advertise to all (clients & non-clients)
– Only the RR needs special config – clients remain unaware of route reflection. This eliminates the need for full iBGP mesh.
PEERING VS. TRANSIT
Transit: ISP provides full reachability (paid relationship)
Peering: ISPs exchange selected routes; equal relationship, usually unpaid
AS Path Filtering: to avoid getting transit; ip as-path access-list 10 permit ^ \$
INTERNET EXCHANGE POINT (IXP)
– Facility where networks exchange traffic via BGP peering
– Reduces transit costs, latency, and offloads upstream links
PUBLIC PEERING
– Members peer via shared switch fabric and a route server
– Route server distributes routes but stays out of data path (NEXT_HOP unchanged)
– Minimal policy control; one BGP session to route server
– Simplified setup (one legal contract)
PRIVATE PEERING
– Direct BGP sessions between two parties (1:1)
– May use public or private interconnects
– Full policy control per neighbor
– Requires one session and legal contract per peer
Examples: Equinix, SwissIX (non-profit)
ENTERPRISE CONNECTIVITY OPTIONS
Single-Homed: One ISP, one link (BGP or static); simple but no redundancy

Dual-Homed: One ISP, two links (or routers); redundancy within same provider
Multihomed: Multiple ISPs; improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes
Dual-Multihomed: Multihomed, but two links per ISP
TRAFFIC ENGINEERING (TE)
Outbound TE (Local Pref): Set higher local pref to prefer exit path; affects outbound traffic (MED);
Inbound TE (Highest Wins): Signal entry preference with MED; lowest wins; only works if peer honors it
Inbound TE (AS-Path Prepending): Add own ASN multiple times on backup path; shortest AS-path wins
TE Limitation: Affects only routes outbound (e.g. local pref); inbound control limited – ISPs may ignore MED
TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for fallover
Aggregate Impact: Longest-match wins → specific prefixes may steer traffic to alternate ISP; avoid provider-owned aggregates
RPKI – RESOURCE PUBLIC KEY INFRASTRUCTURE
– Framework to validate that a prefix is legitimately originated by a specific ASN
– Prevents route hijacking and accidental mis-originations (route leaks)
– RPKI validates origin → not the 5 RRs
KEY COMPONENTS
Trust Anchors (TA): Root CAs of the AS-Paths; issue certs for resource holders
ROA – Route Origin Authorization: Digitally signed object that authorizes ASN to announce a prefix (contains AS, prefix that AS can originate, max prefix length)
RPKI Validators: Software that downloads, verifies, and stores Validated ROA Validations (VRPs)
RPKI VALIDATION STATES
Valid: Prefix + ASN match ROA
Invalid: Prefix found, but ASN mismatch or prefix too long
Not Found / Unknown: No matching ROA
VALIDATOR DETAILS
– Use TALS (Trust Anchor Locators) to fetch data from RIRs
– Validate cryptographic signatures (via X.509 certs with RFC 3779)
– Outputs VRPs; invalid objects are discarded
– Update frequency: ~24h (recommended), ~30–60min (practice)
– RRDP (RFC 8182) replacing rsync (uses HTTPS)
BGP MONITORING
Event tracking, BGP hack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization
NETWORK DESIGN
Pillars: Scalability, Speed, Availability, Security, Manageability → overall Cost
AVAILABILITY CONCEPTS (MOST IMPORTANT REQUIREMENT)
MTBF: Mean Time Between Failures; MTTR: Mean Time to Repair
$A = \frac{MTBF}{MTBF + MTTR}$ MTBF combined: $\left(\sum_{n=1}^N \frac{MTBF_n}{MTBF_n + MTTR_n} \right)^{-1}$ parallel; $\frac{MTBF_n}{n}$
– Lower MTTR + higher MTBF = more availability
REDUNDANCY
– Adds reliability, decreases MTBF but increases MTTR and complexity
– Balance: resilience vs. manageability
Backup Paths: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
Load Balancing: ECMP, EtherChannel, Port-Channel
HIERARCHICAL DESIGN
Access: Connect end devices, high port count, port security, L2, QoS marking
Distribution: L3, policy control, HSRP/VRP, loop protection, small fault domain
Core: High-speed backbone, L3 only, no policy, scalable/redundant, no security
Collapsed Core: Combines Core + Distribution (small/medium networks)
FLAT DESIGN
– Any-to-any; small networks, MPLS/LAN setups
– Lacks scalability and control
FABRIC DESIGN
Modern design using Underlay/Overlay
Underlay: transport (e.g., IP, MPLS)
Overlay: logical virtual topology (e.g., EVPN)
ENTERPRISE CAMPUS
– 100s/1000s of users, multiple buildings, one physical location
– Multiple interconnected LANs, connected via Ethernet and Wireless
FHRP – FIRST-HOP REDUNDANCY PROTOCOLS
– Ensures default gateway is always reachable, PCs can only have one
– Enables fast failover during router failure
VRPP – Virtual Router Redundancy Protocol (Multivendor): Shared virtual IP, real MACs per router, Master router handles forwarding
HSRP – Hot Standby Router Protocol (Cisco): Shared virtual IP + virtual MAC, One active, others in standby → faster
GLBP – Gateway Load Balancing Protocol (Cisco): Load balancing + redundancy, Shared virtual IP + multiple virtual MACs, Roles: AVG: Answers ARP and sends virtual MAC addresses of AVFs AVF: Forwards traffic
DATA CENTER DESIGNS
ToR – Top of Rack: switches per rack, less cabling, easy expansions/ exchanges per 'rack', scalable glass fiber, ideal for high service density (full racks). But more switches, more ports, more L2 Srv-2 Srv traffic, more STP to be managed
EoR – End of Row: 1 switch per row, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. But more cabling
TRAFFIC DIRECTIONS
North-South: Between external networks (client-server, in/out DC)
East-West: Within DC (e.g., server-server, storage)
THREE-TIER DC
– Access – Aggregation – Core (like Hierarchical)
– Optimized for North–South traffic
– Not ideal for East–West communication
LEAF-SPINE ARCHITECTURE
– Two-tier: Leaf switches (access) connect to Spines (core)
– High performance, low latency
– Scalable, ideal for East–West traffic

MULTICAST
Use-cases 1-to-Many: Streaming, software updates, music-on-hold
Use-cases Many-to-Many: Gaming, VR, stock data, group chat
Benefits: Efficient bandwidth, lower server/CPU load, no redundancy, supports multipoit apps
Properties: UDP-based (no delivery guarantee, congestion control, or ordering) Apps must handle drops, duplicates, out-of-order packets
Source: Sends to group IP; doesn't need to join
Receiver: Must explicitly join group to receive traffic
MULTICAST ADDRESS RANGES
224.0.0.0 – 224.0.0.255: Link-local, TTL = 1 (not forwarded by routers)
224.0.1.0 – 224.0.1.255: Reserved by IANA, routable
224.0.0.0 – 239.255.255.255: Source-Specific Multicast (SSM)
239.0.0.0 – 239.255.255.255: Administratively scoped (private multicast space)
BROADCAST BASICS
L3 routes between subnets; L2 floods within same subnet
L2 (Bridging): MAC flit.filt.filt; switches flood to all ports in VLAN
L3 (Routing):
– 255.255.255.255: local broadcast, not routed
– Directed broadcast (e.g. 10.1.1.255): can be routed if enabled
L2 MULTICAST: MAC MAPPING
1) Get IP: Example multicast IP address: 239.5.5.5
2) Convert to Binary: 239.5.5.5 = 11011111.00000101.00000101.00000101
– Take only last 23 bits: 00000101.00000101.00000101
3) Map to MAC Prefix: Use fixed MAC multicast prefix: 01:00:5E → Map last 23 bits to: 01:00:5E:05:05:05
4) Final MAC Address: 01:00:5E:05:05:05
IGMP – INTERNET GROUP MANAGEMENT PROTOCOL (HOST TO FIRST-HOP-ROUTER)
Purpose: Manages group membership for IPv4 multicast on each segment

IGMPv1:
– Basic join via query-response mechanism
– No way for a host to leave a group explicitly
– Router sends general membership queries every 60s to 224.0.0.1
– If no report is received, router removes group after timeout
– Receiver has no knowledge of the multicast source
IGMPv2:
– Adds Leave Group message (faster pruning of unused traffic)
– General queries to 224.0.0.1 every 125s 'Any hosts interested in any groups?'
– Supports Group-Specific Queries e.g. when someone leaves group ('anyone still interested in group x?'), reducing broadcast overhead
– Still source-agnostic: receivers don't know who the source is

IGMPv3:
– Adds source filtering (Include/Exclude lists)
– Enables Source-Specific Multicast (SSM) – receiver requests traffic only from selected source(s), no need for Rendezvous Point (RP) anymore
– Adds support for applying source-level access control (source filtering)
– Can also be used in ASM (Any Source Multicast), but mainly with SSM

IGMP SNOOPING
Without snooping: multicast = broadcast on VLAN
With snooping: switch listens to IGMP messages and builds a forwarding table
Default: snooping is enabled; switch needs IGMP Query to operate
REVERSE PATH FORWARDING (RPF)
RPF Check: To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of.

RENDEZVOUS POINT (RP)
– Used in Shared Tree (*,G) setups with PIM-SM
– RP acts as the common meeting point for sources and receivers
– RPF check is performed toward the RP (not the source)
– Once the source is known, routers may switch to a Source Tree (S,G)
SHARED TREE VS. SOURCE-BASED TREE
Shared Tree (*,G):
– IGMP host sends a membership report (IGMP Join)
– Router adds (*,G) entry to multicast routing table
– * means 'any source' – source is unknown/unspecified
Source-Based Tree (S,G):
– Built when router receives an (S,G) join/report from IGMP host
– S = known multicast source; G = group
– Router adds (S,G) to mroute table once source is known

PIM – PROTOCOL INDEPENDENT MULTICAST (ONLY BETWEEN ROUTERS)
– Relies entirely on the unicast routing table (RIB) for multicast forwarding decisions
– Protocol-independent: works with static routes, OSPF, IS-IS, etc.
PIM-DH – DENSE MODE (NO RP)
Push Model: Floods multicast traffic to all interfaces; then prunes where no receivers exist

1) Flooding: Source sends traffic → forwarded out all multicast-enabled links using unicast RIB
2) Distribution Tree: Initially includes entire network (shared tree rooted at source)
3) Prune Messages: Routers without interested receivers send prunes upstream to remove themselves from the tree
4) State Maintenance: Routers track source, receivers, interfaces to forward/prune per group
PIM-SM – SPARSE MODE (ASM, SSM)
Pull Model: Multicast traffic is only sent where requested. Works with IGMP to detect interested receivers and uses unicast routing for forwarding.

1) Join/PRun: Routers send explicit join/prune messages to request or stop receiving multicast for group (G) to other routers
2) Forwarding: Routers only forward multicast packets for group (G) on interfaces from which explicit joins were received
ASM – ANY SOURCE MULTICAST (HOW IT WORKS WITH RP)
– Works with IGMPv1 or IGMPv2 → receiver does not know the source
– Receiver sends IGMP Join (*,G) to its first-hop router
– First-hop router forwards PIM Join (*,G) hop-by-hop toward the Rendezvous Point (RP)
– RP acts as a common meeting point for sources and receivers
– Sources send multicast traffic to the RP via a PIM Register tunnel
– All routers in the multicast domain must know the RP location

Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.

WAN
Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
Bandwidth: App needs, peak usage, reserve for VoIP
Control/Security: Trust provider? No full control
Availability: Redundancy, SLA for failures

Global segments: Defined in the SR Global Block (SRGB) and should be consistent across all nodes; **local segments** are defined in the SR Local Block (SRLB) and are specific to the local SR node.

KEY BENEFITS

- Latency Reduction:** Nearby edge servers reduce round-trip time
- Availability:** Failover and redundancy in case of node failure
- Scalability:** Handles traffic spikes via load balancing
- Cost Optimization:** Reduces backend and transit load on origin
- DDoS Protection:** Edge servers absorb attacks → not all traffic on one server

```

  origin IGP, MED not set, localpref 100, weight 0
  received label 30010 50000
  extcommunity: RT:1:10 RT:65000:50000 ENCAP:8 Router MAC:5254.00ca.69ae
  originator: 172.16.255.101 Cluster list: 172.16.255.1

```